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Computation & theory

Machine learning-assisted design of biomedical high entropy alloys with low elastic modulus for orthopedic implants

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ABSTRACT

This paper focuses on finding an optimum composition for the TiTaHfNbZr quinary high entropy alloy (HEA) system with an elastic modulus close to that of bone in order to attain a better biomechanical compatibility between the bone and the implant in orthopedic applications. To obtain the composition providing the desired structural match, machine learning (ML) tools were implemented in the current work instead of conventional trial-and-error methods. The ML algorithms utilized in this study were trained using experimental data available in the literature and then utilized to predict the optimum HEA compositions with the lowest elastic moduli. Consequently, the $Ti_{23}Ta_{10}Hf_{27}Nb_{12}Zr_{28}$ and $Ti_{28}Ta_{10}Hf_{30}Nb_{14}Zr_{18}$ compositions were predicted as the optimum HEA compositions with elastic moduli of 83.5 ± 2.9 and 87.4 ± 2.2 GPa, respectively. The materials were manufactured, and the elastic moduli were validated with nanoindentation experiments. The samples were also exposed to static immersion experiments in simulated body fluid (SBF) for 28 days to gain insight and information regarding the ion release and ensure that the new HEAs are biocompatible. The findings of the work reported herein demonstrate that the proposed ML model can successfully predict HEA compositions for an optimized biomechanical compatibility for orthopedic applications and warrant further biomedical research on the two new HEAs prior to their utility as orthopedic implant materials.

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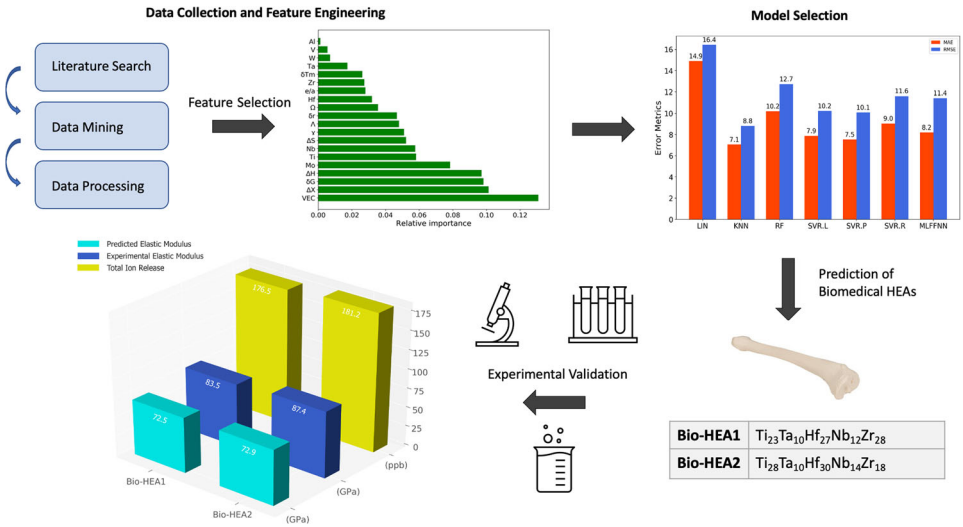
GRAPHICAL ABSTRACT



Introduction

Metallic materials, being utilized in the manufacturing of more than 70% of medical implants, have a substantial standing in biomedical applications since they are substituted for failed hard tissue in human body to enhance the quality of life of the patients [1–3]. Compared to other metallic biomaterials such as Co-based alloys or stainless steels, Ti and its alloys have gained popularity in medical and dental applications owing to their excellent biocompatibility and corrosion resistance [4–6]. Recently, Ti-based high entropy alloys (HEAs), comprising biocompatible elements such as Ta, Hf, Nb, Zr and Mo, have been studied by researchers as potential alternative implant materials [7–10]. As opposed to conventional metallic implant materials comprised of one or two principle elements along with minor additional elements, the HEAs are composed of 4 or more principal elements with concentrations varying between 5 and 35 at% in equimolar or near-equimolar proportions [11, 12]. Due to the high configurational entropy brought about mixing of multiple principal elements promoting the formation of a single stable solid solution phase [8, 13], HEAs exhibit superior chemical and mechanical properties such as excellent corrosion resistance [14, 15], high hardness and wear resistance [16–18], high strength and toughness [8, 15] and high thermal stability at elevated temperatures [15, 19]. For the last 10 years, promising novel biocompatible HEAs have been synthesized, and it has been confirmed that these alloys have a favorable combination of physical and mechanical properties in comparison to other biomaterials [9]. Even though some biocompatible HEAs possess lower elastic moduli than currently implemented biomedical alloys such as 316L, CoCrMo, and Ti6Al4V [9], they are still far from the elastic modulus of bone, which varies between 4 and 30 GPa [3, 20]. Therefore, the long-term functionality of these implants decreases due to structural dissimilarities.

The stiffness mismatch between the implant and the bone promotes possible implant loosening, failure of the implant, and tissue loss [6, 8, 21]. Namely, the implant material whose elastic modulus is much higher than that of the bone starts to carry the majority of the applied stresses, leading to inhomogeneous allocation of the exerted load between the implant and the surrounding tissue. As a result of this phenomenon also known as stress shielding, the bone adapts itself by either reducing its mass (bone absorption) or becoming thinner (bone remodeling) [1, 2, 5, 6, 22]. The aforementioned challenges bring

about a need for the search for new implant materials with a desirable combination of biological and biomechanical compatibilities [2, 5, 6]. In particular, it is essential for the implant to be strong and durable with a structural stiffness matching to that of the bone in order for it to properly function in the long term without failing or requiring revision surgery [1–3]. Hence, the investigation and optimization of biocompatibility of HEAs as potential implant materials from both biological and mechanical aspects are crucial for their safe utility as implant materials. Such examples include equimolar TiZrNbTaMo HEA possessing an elastic modulus of 153 GPa, an excellent corrosion resistance and high stability against pitting [23], as well as equimolar TiZrTaHfNb and $Ti_{1.5}ZrTa_{0.5}Hf_{0.5}Nb_{0.5}$ possessing elastic moduli of 112.74 and 98.57 GPa, respectively, with excellent corrosion resistance, good wettability and wear resistance [9]. The variations in the elastic moduli of the HEAs with the same constituent elements [9] evidence that alloying elements in different quantities alter the properties of HEAs substantially, making new alloy design more attractive and broadening their utility for different applications [2, 24].

Alloy design with preferred intrinsic properties can be accomplished through controlled chemical alloying which modifies microstructure and phases present in the matrix [2, 13, 25]. Sakaguchi et al. explored ternary and quaternary TiNbTaZr system with varying Nb (0–40 wt.%), Ta (0–20 wt.%) and Zr (0–10 wt.%) contents. It was demonstrated that Nb and Ta concentrations affect the stress–strain behavior; therefore, the deformation mechanisms and the mechanical properties. For instance, elastic modulus of the system first decreases and then increases as Ta concentration changes from 0 to 20 wt.%, validating high sensitivity of the system's intrinsic properties to Ta content. Furthermore, varying Nb concentration results in a more complicated modification in elastic modulus as compared to the effects of Ta and Zr concentrations [26–28]. Additionally, based on the investigations on TiV and TiNbSn alloys [29], and Ti35Nb4Sn6Mo_xZr (x varying from 0 to 15 wt.%) system [5], it was shown that the elastic modulus depends on chemical composition in an inexplicit way. Thus, considering the design of an alloy with more than 4 principal elements such as HEAs, determining the interactions and resulting properties by trial-and-error or intuition remains an inefficient approach as there exists a very large search space for HEAs with a high number of possible combinations of the constituent elements [13, 14, 30].

An effective and very efficient remedy for this problem is the implementation of machine learning (ML), which has recently become popular in materials science community. For the last couple of years, increase in the availability of data and the ML methods' ability to identify underlying trends from data have expedited materials research and discovery by significantly decreasing computational and experimental costs as compared to traditional methods [31–36]. Specifically, the ML algorithms are trained using a training dataset, validated and then utilized to make predictions on the test dataset which are not provided to the ML algorithms during the initial training and validation processes [37, 38]. Such an approach is very advantageous in materials discovery where the compositional search space is very large and the relations among parameters are unknown [35, 36]. In recent years, ML methods have been put into use in various fields of materials science, including phase selection in HEAs [39–45], hardness prediction [15, 46], corrosion resistance optimization [47, 48], improvement of properties of shape memory alloys (SMAs) [49–51], and molecular dynamics simulation integrated property predictions [52, 53].

The aim of the current work is to obtain the optimum composition for $Ti_xTa_yHf_zNb_mZr_n$ biomedical HEAs with elastic moduli close to the stiffness of the bone by employing ML algorithms with the specific purpose of providing a better structural match between the implant and the bone in orthopedic applications. To reach this target, first, a new dataset was curated from the existing literature on Ti-containing HEAs and medium entropy alloys (MEAs) along with their measured elastic moduli. Then, the dataset was used to train several ML algorithms which were employed to construct a model to predict the elastic modulus of new compositions. In the final step, two of the predicted biomedical HEAs with optimum compositions were subjected to validation experiments to confirm the accuracy of the predictions. As a result, two novel biomedical HEAs, namely $Ti_{23}Ta_{10}Hf_{27}Nb_{12}Zr_{28}$ and $Ti_{28}Ta_{10}Hf_{30}Nb_{14}Zr_{18}$, were obtained with elastic moduli of 83.5 ± 2.9 and 87.4 ± 2.2 GPa, respectively. The work presented herein, thus, illustrates that the ML methods were successfully implemented to optimize $Ti_xTa_yHf_zNb_mZr_n$ biomedical HEA compositions with the aim

of minimizing elastic modulus in order to ensure mechanical compatibility between the implant and bone.

Computational methods

Data mining

To construct the dataset required for training of the ML algorithms, first, a detailed literature search was conducted within the HEA literature published since 2000s. Then, all the articles on quaternary, quinary, and senary as-cast HEAs synthesized by arc melting process including the experimentally measured elastic moduli information were scanned and put together to form the preliminary dataset. Then, the elastic moduli of datapoints with the same compositions were averaged and treated as a single datapoint. The dataset used to train the ML algorithms and make predictions consisted of 53 datapoints, which is provided in Supplementary Information Table S1 (each datapoint, i.e., each row in the dataset, has alloy composition, calculated empirical parameters and corresponding elastic modulus as target variable). In order to increase the number of datapoints in the training dataset and enhance the generalization capability of the ML models [46], HEAs containing toxic elements such as V and Al were also considered. The dataset utilized in this work possesses a limited number of datapoints for the following reasons: The occurrence of the published articles in the literature on biomedical HEAs has not been very frequent until recent years, and not all the published literature has the relevant experimental data. In fact, acquisition of large numbers of datapoints is one of the main struggles in materials informatics where obtaining materials and corresponding target variables are experimentally challenging and costly [35], leading to datasets with limited amount of datapoints.

Each datapoint in the dataset is composed of nine elements (Ti, Ta, Hf, Nb, Zr, Mo, W, V, Al), eleven empirical features, and one target variable, i.e., the elastic modulus. The 11 calculated empirical features are valance electron concentration (VEC), atomic size difference (δr), difference of electronegativity ($\Delta\chi$), mixing enthalpy (ΔH), configurational entropy (ΔS) [54], Ω parameter proposed by Yang et al. [55], Λ parameter defined by Singh et al. [56], γ parameter suggested by Wang et al. [57], number of itinerant electrons (e/a) [58], difference of shear modulus (δG), and difference of melting temperature (δT_m). The latter two empirical features are proposed based on the analogy to δr. Most of the aforementioned empirical features are related to crystal structure, phase type and formation or stability of single-phase solid solution of HEAs, which are the governing factors for controlling mechanical properties in HEAs [13, 59]. All the empirical features utilized herein are listed in Table 1.

In Table 1, the features C_i, $\mathrm{VEC_i}$, $\mathrm{r_i}$, χ_i, $(e/a)_i$, G_i, and T_i refer to the atomic concentration, VEC, atomic radius, Pauling electronegativity, e/a, shear modulus, and melting temperature of each element in an alloy, respectively. The averages of atomic radius$\overline{r}$, Pauling electronegativity$\overline{\chi}$, shear modulus $\overline{G}$, and melting temperature $\overline{T}$ are calculated as$\overline{r} = \sum_{i=1}^{n} C_i r_i$, $\overline{\chi} = \sum_{i=1}^{n} C_i \chi_i$, $\overline{G} = \sum_{i=1}^{n} C_i G_i$, and$\overline{T} = \sum_{i=1}^{n} C_i T_i$, respectively. R, n, $r_{\min}$, and $r_{\max}$ are universal gas constant, the number of constituent elements, and the atomic radii for the smallest and largest atoms, respectively. $\Delta H_{ij}^{\mathrm{mix}}$ is the mixing enthalpy of atomic pairs in a binary system estimated by Takeuchi and Inoue utilizing Miedema method [60].

In ML applications, due to the nature of the predictors within the dataset, a preprocessing of the variables may be required to improve the performance of the whole ML process. Specifically, the variables in a dataset might have a non-normal distribution and can be highly skewed or in different scales, all of which deteriorate the general performance of the ML algorithms. Therefore, an analysis of the dataset should be carried out before training to understand the data and implement the appropriate data transformation process. With data transformation, the variables are normalized and scaled so that all of them are treated equally by the ML algorithms [39, 48]. In this study, a min–max normalization method was applied to scale and normalize the features between 0 and 1 (Supplementary Information, Equation S1).

Feature selection

One of the major difficulties in ML implementations with a high-dimensional dataset is selection of the most relevant features to train the ML algorithms. Too many input features result in overfitting,

Table 1 List of empirical features and the corresponding mathematical formulae

Description of empirical features	Formulae
Valance electron concentration (VEC)	$\mathrm{VEC} = \sum_{i=1}^{n} C_i * \mathrm{VEC}_i$
Atomic size difference (δr)	$\delta r = \sqrt{\sum_{i=1}^{n} C_i * (1 - \frac{r_i}{\bar{r}})^2}$
Difference of electronegativity ($\Delta\chi$)	$\Delta\chi = \sqrt{\sum_{i=1}^{n} C_i * (\bar{\chi} - \chi_i)^2}$
Mixing enthalpy (ΔH)	$\Delta H = \sum_{i=1,j>1}^{n} 4 * C_i * C_j * \Delta H_{ij}^{mix}$
Configurational entropy (ΔS)	$\Delta S = -R \sum_{i=1}^{n} C_i * \ln(C_i)$
Ω parameter	$\Omega = \frac{\bar{T} * \Delta S}{\Delta H}$
Λ parameter	$\Lambda = \frac{\Delta S}{\delta r * \delta r}$
γ parameter	$\gamma = \left(1 - \sqrt{\frac{(\bar{r}+r_{\min})^2 - \bar{r}^2}{(\bar{r}+r_{\min})^2}}\right) / \left(1 - \sqrt{\frac{(\bar{r}+r_{\max})^2 - \bar{r}^2}{(\bar{r}+r_{\max})^2}}\right)$
Number of itinerant electrons (e/a)	$e/a = \sum_{i=1}^{n} C_i * (e/a)_i$
Difference of shear modulus (δG)	$\delta G = \sqrt{\sum_{i=1}^{n} C_i * (1 - \frac{G_i}{\bar{G}})^2}$
Difference of melting temperature (δT_m)	$\delta T_m = \sqrt{\sum_{i=1}^{n} C_i * (1 - \frac{T_i}{\bar{T}})^2}$

especially in datasets containing small number of instances, due to high degrees of freedom, they introduce into the algorithms, leading to a decrease in the generalization capability of the ML models. Hence, a careful selection of features which explain most of the variance in a target variable is required to discard the redundant information from a dataset, and thus, has a critical impact on the prediction accuracy and performance of a model [47]. Choosing the best set of features becomes an even more difficult task in the presence of high correlation among the features. Highly correlated features affect model performance and prediction accuracy due to the fact that they have the same information about the target variable, i.e., the essence of the correlated features is so similar, such that the ML algorithms cannot differentiate among them [35, 61]. One of the frequently used parameters for the correlation analysis is the Pearson correlation coefficient (PCC) [37] (Supplementary Information, Equation S2), but it fails to capture the nonlinearity between the features. To overcome this issue, random forest (RF) method which is a very attractive choice for classification and regression problem was also employed for feature selection in this work. Compared to other feature selection methods, the RF method provides a measure of importance of variables by ranking them based on their performances during the optimization of the model [61, 62]. Thus, by combining the two methods, the most relevant features were selected to train the ML algorithms.

Machine learning algorithms and synthetic dataset

In this study, different ML algorithms, namely linear regression (LIN), *k*-nearest neighbors regression (KNN), RF, support vector regression with a linear kernel (SVR.L), support vector regression with a polynomial kernel (SVR.P), support vector regression with a radial basis kernel (SVR.R), and multi-layer feed-forward neural network (MLFFNN), were evaluated based on their performance in predicting the target variable. Prior to training the models, the initial dataset was divided into a training (used for optimization of the models) and test dataset (used at the end to obtain the models' performance accuracy predicting the unseen data). While optimizing the models using hyperparameters found with grid search strategy, a method in which various combinations of hyperparameters are tried out to select the best batch, repeated holdout validation and bootstrapping with replacement were utilized to improve training process to avoid overfitting as much as possible with a small training dataset. The ML models were evaluated based on their prediction errors on the test dataset. The error metrics utilized are root mean squared error (RMSE) and mean absolute error (MAE) (Supplementary Information, Equations S3 and S4). All computations were performed using scikit-learn library with Python programming language [63]. To find the optimum composition leading to the smallest possible elastic modulus, a synthetic dataset was constructed with

more than 500000 different compositions, where Ti, Ta, Hf, Nb, and Zr contents ranged from 5 to 35 with 1 at% intervals, complying with the widely accepted HEA definition. Then, the synthetic dataset was fed into the best ML model to make new predictions in the aforementioned composition range.

Computational results

The effect of training and test set split proportions

The general notion in any ML algorithm is that the larger the dataset, the better the model performance will be, because a large dataset contains an extensive amount of information about the target variable. Therefore, both training and test datasets randomly taken from this large dataset will have a normal distribution representing the whole sample space for both sets, making any model better at generalizing and reducing variance due to a reduced outlier effect. Overfitting due to outliers can be alleviated with data transformation such as normalizing and scaling of the data [47]. However, when the dataset is small, normalization and scaling of the data might not be as effective as it is in the case of a large dataset. In addition to a transformation, a careful split of data creating a bell-shaped distribution of both training and test datasets should be taken into consideration to avoid any overfitting. Therefore, the ratio of split of a small dataset similar to the one employed in this study is very critical so as to make sure that datapoints in both sets follow the same distribution. As illustrated in Fig. 1a, the prediction error for KNN, SVR.R, SVR.P, SVR.L, and RF decreases as the proportion of training dataset increased from 30 to 90%. The RMSE values shown in Fig. 1a represent the mean value of RMSEs estimated 200 times on randomly selected testing set for different algorithms by using only the compositions as input variables. As shown in Fig. 1b, bootstrapping with replacement was also implemented as another evaluation method for RMSE using 70% of the training dataset. A split ratio of 70% training dataset was considered in this work because of the limited number of datapoints present in the dataset. Decreasing the proportion of the datapoints in the testing set may be misleading in the evaluation of the models as only few datapoints in the testing set will contribute to the calculation of RMSE. Based on Fig. 1, KNN algorithm outperforms others and thus was selected as the main model in this work.

Feature selection

The aim of selecting the right features is to eliminate any redundant or duplicate information present in the dataset and decrease degrees of freedom of the models so as to reduce complexity and prevent overfitting [61]. The correlation of each input feature to the target, i.e., elastic modulus, is presented in Supplementary Information Fig. S1. Solely based on Fig. S1, δr, Al, and δT_m features seem to be very uncorrelated with elastic modulus, indicating that they do not contain relevant information about the target. The PCC values for the empirical features δG, $\Delta\chi$, and VEC, on the other hand, are larger than 0.5, indicating a strong correlation between these three empirical features and the target. While

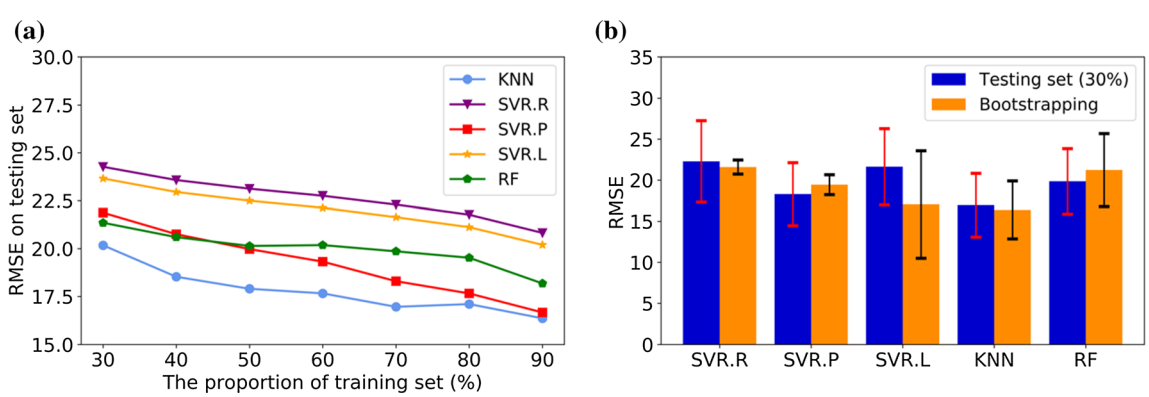


Figure 1 **a** The evolution of RMSE on test dataset with changing training dataset proportions for KNN, SVR.R, SVR.P, SVR.L, and RF to evaluate and select the best performing model. **b** Estimation of RMSE with repeated holdout validation approach and bootstrapping methods, pointing out KNN is the best performing model.

understanding the relationship between input and output variables is helpful in conceptualizing a preliminary plan for feature selection, the investigation of the correlation among input features is also crucial as the correlation between them deteriorates efficiency and accuracy of ML algorithms [64].

The heatmap for PCCs among input variables is presented in Fig. 2. Accordingly, Mo and $\Delta\chi$ have a PCC of 0.95, which is the highest correlation among all features. The second highest correlated features are Mo and δG, and Al and ΔH. In an ideal scenario, all input variables should be uncorrelated [35]. Therefore, only one feature is selected in the presence of strong correlation between features to prevent duplicate information in the dataset [15, 47]. There are two main drawbacks of PCC: First, it neglects any existing nonlinearity in the dataset; second, the performance of ML algorithms is not included in the process of estimating PCCs [44, 61].

To decrease the effect of high correlation in the dataset, the RF method was employed in the process of feature selection in addition to PCC. RF consists of a forest with many decision trees, each of which is created by only considering a subset of features resulting in uncorrelated trees as compared to basic decision trees. One advantage of RF is that it provides the user with the feature importance making it an attractive tool for feature selection. By restricting each tree to consider only a subset of all available features, it calculates each feature's performance/importance in increasing the accuracy when splitting a node while growing a tree [65]. The corresponding feature importance results obtained from the RF algorithm utilized in this work are demonstrated in Fig. 3.

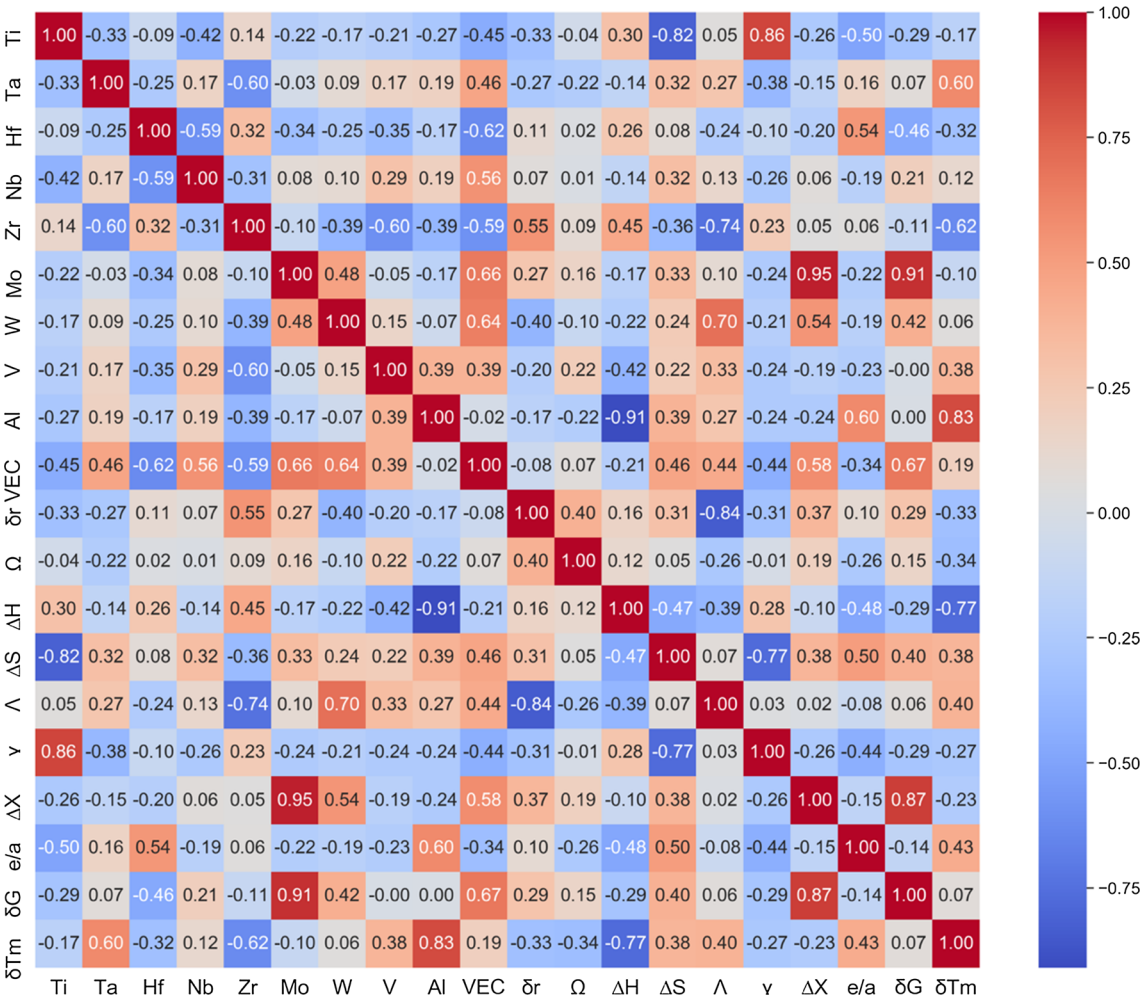


Figure 2 The heatmap of PCCs for chemical composition and empirical features. The higher the color intensities are, the more the correlation between the two features is. Red and blue colors indicate positive and negative correlation, respectively.

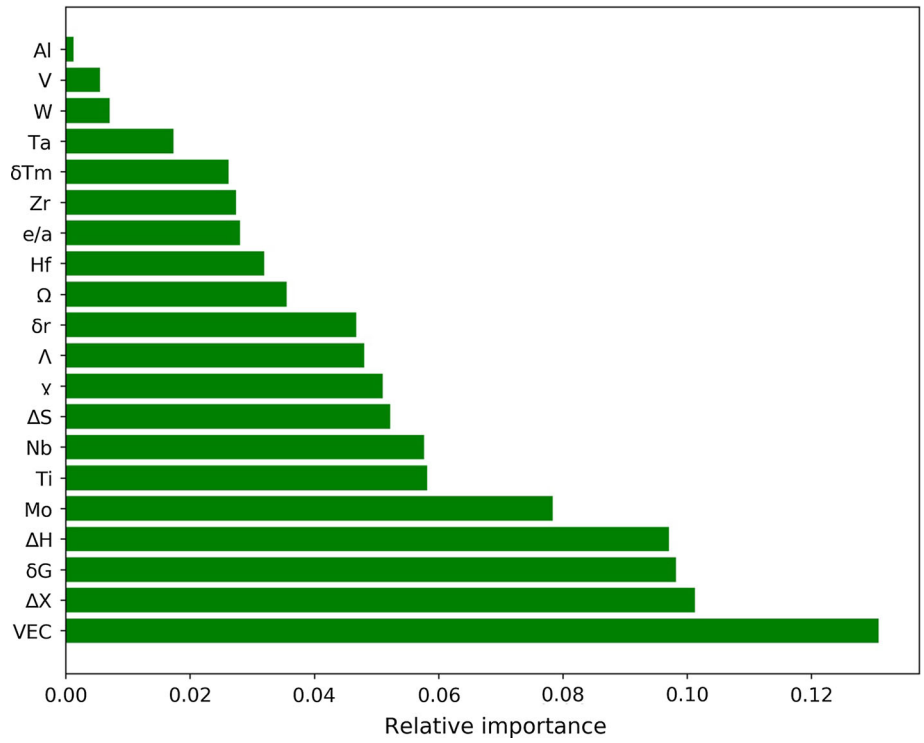


Figure 3 The relative feature importance for input variables based on their performance in reducing error while splitting a node when constructing trees in RF method.

Accordingly, VEC stands out as the best predictor, leaving all other compositional and empirical features behind. The next three most important features are $\Delta\chi$, δG, and ΔH, suggesting that empirical features contain relevant information related to intrinsic properties of the alloys within the dataset. Compared to Fig. S1, W does not seem to be of importance when the model performance is included in the selection process. Based on Fig. S1, ΔH is not highly correlated with the target. On the other hand, when its performance is evaluated considering its importance as a feature in RF method as shown in Fig. 3, it can be inferred that elastic modulus is nonlinearly dependent on ΔH. Since Mo and $\Delta\chi$ are highly correlated and Mo is one of the elements representing alloys in the dataset, $\Delta\chi$ was not considered in the following feature selection process.

The total number of empirical features to be included in ML algorithms leading to minimum prediction error was investigated iteratively using recursive feature elimination method based on RF algorithm [61]. Figure 4 presents the evolution of RMSE of the RF method on the test dataset by iteratively adding one feature at a time on top of each other, greedily selecting the best performing feature and eliminating the rest at each iteration. In Fig. 4, the green circle which represents the subset

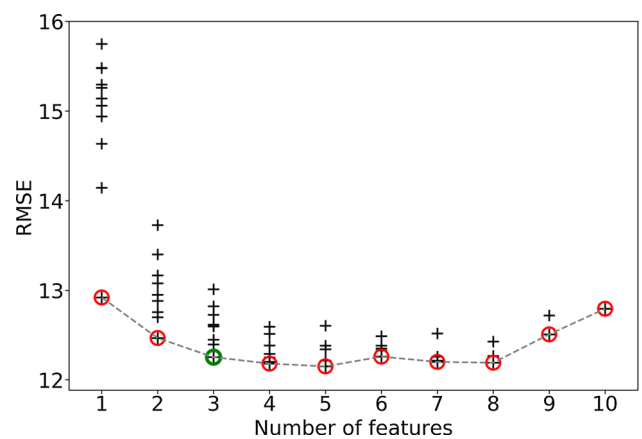


Figure 4 Variation of RMSE with the addition of each empirical parameter on composition utilizing recursive feature elimination method combined with the RF model. The red circles show the best performing subset of features resulting in the least prediction error for each combination of features. The green circle indicates the selected subset of features as a result of feature elimination which are ΔH, VEC, and δG.

consisting of three features, namely ΔH, VEC, and δG, was selected, although the minimum RMSE was obtained with the addition of the fifth empirical feature since there is no substantial improvement in RMSE after the addition of the fourth feature. Therefore, the final set of features included the chemical compositions and the three empirical

features, namely ΔH, VEC, and δG, in order for the models not to lose their generalization capabilities. Shaikh et al. reported that increase in VEC enhances average bond strength leading to a higher elastic modulus, as well as melting temperature and density of HEAs, which indicates a positive correlation between VEC and elastic modulus [66]. Furthermore, VEC was also utilized as a parameter for designing ductile HEAs where it was reduced through controlled alloying [25, 67]. Furthermore, it was reported that $\Delta H_{ij}^{\mathrm{mix}}$ is related to bond strength, and as the number of bonds with high $\Delta H_{ij}^{\mathrm{mix}}$ rises, elastic modulus increases [68]. Therefore, ΔH represents the information inherently at the atomistic level. Finally, Fig. 4 indicates that utilizing composition with empirical features associated with HEAs in ML framework augments the learning process by extracting more information from the training dataset [15].

Machine learning results

Various ML algorithms were trained using composition and three selected features and employed to predict novel biomedical HEAs with low elastic modulus. RMSE and MAE values on the test dataset for seven different ML methods, namely LIN, KNN, RF, SVR.L, SVR.P, SVR.R, and MLFFNN, are presented in Fig. 5. Accordingly, the worst-performing ML algorithm was LIN as compared to others. The underlying reason of this relatively poor performance might be the fact that the relationship between input and output variables is highly complex and nonlinear so that a linear model is not an appropriate choice to extract information from the dataset. On the other hand, RF, SVR.L, SVR.P, SVR.R, and MLFFNN showed almost the same performance in terms of their prediction capabilities, although RMSE and MAE values for RF were the highest among the five models. Ultimately, the best performing model was the KNN method with the least RMSE and MAE values. Figure 5 demonstrates that utilization of the three chosen features in addition to chemical composition in the training of the ML algorithms resulted in an improvement of more than 50% in RMSE for RF, SVR.L, SVR.P, SVR.R, and KNN models (Fig. 1). This shows the importance of using additional empirical features along with compositions in accelerating the optimization process of new HEAs.

The KNN algorithm can be utilized for both classification and regression problems. This method utilizes the training dataset as a guide to make new predictions based on resemblance of datapoints. Predictions for datapoints are made considering how close they are to a number of datapoints in training dataset using a distance metric. In the KNN method, the k parameter defines the number of the closest neighbors in training dataset when a new datapoint is predicted. When k is equal to 1, the method is highly flexible closely following any pattern in training dataset. As the k parameter increases, the model grows to be less flexible with some generalization capabilities. The algorithm can assign weights equally or using a distance function between neighbors and datapoints. In this study, the hyperparameters implemented for training the KNN algorithm

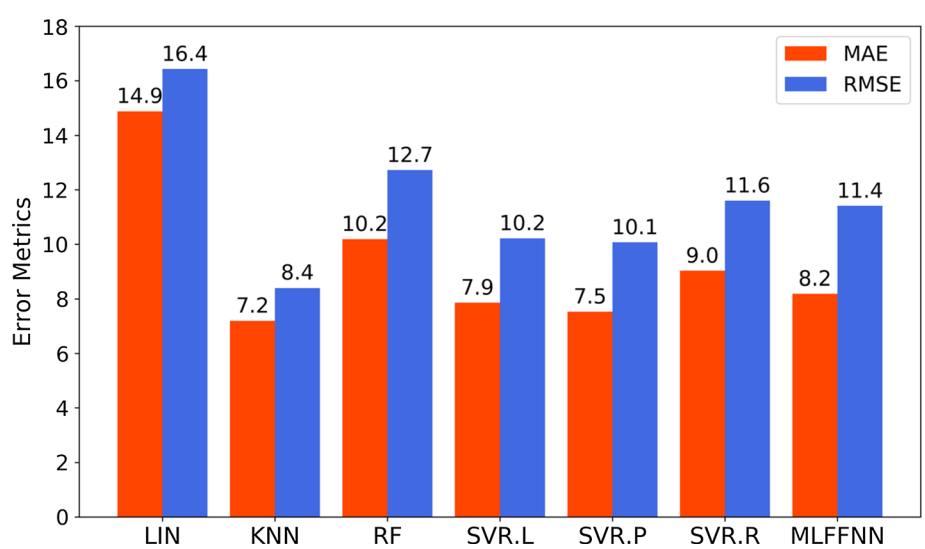


Figure 5 RMSE and MAE on test errors for various ML algorithms employed in this study by using composition and three selected features as input.

were as follows: *k*, *weights*, and *metric* which were set as '2,' 'uniform,' and 'manhattan,' respectively. The variation in RMSE value with respect to growing *k* hyperparameter and the true vs predicted elastic modulus values for the datapoints in the test dataset are demonstrated in Fig. 6a. Figure 6b, c and d illustrate the actual and predicted elastic moduli for KNN, SVR.P, and SVR.L, respectively. Figure 6b clearly demonstrates that the prediction performance of the KNN algorithm was better than that of SVR.P, and SVR.L. Next, the synthetic dataset with more than 500000 different compositions with Ti, Ta, Hf, Nb, and Zr contents ranging from 5 to 35 at% with 0.1% intervals, complying with the widely accepted HEA definition, was imported to the KNN model to make new predictions in the aforementioned composition space for each datapoint. The two optimum compositions having the smallest elastic modulus predictions, namely 72.5 and 72.9 GPa, were selected for validation experiments. The compositions, the predicted elastic moduli and the empirical features for selected alloys are presented in Table 2.

Table 2 Composition and empirical feature values for selected alloys

Selected alloys	Bio-HEA1	Bio-HEA2
Ti (at%)	23	28
Ta (at%)	10	10
Hf (at%)	27	30
Nb (at%)	12	14
Zr (at%)	28	18
VEC	4.22	4.24
δr (%)	4.81	4.73
$\Delta\chi$	0.12	0.12
ΔH (kJ/mol)	2.03	2.08
ΔS (kJ/mol K)	12.74	12.74
Ω	14.94	14.67
Λ	0.55	0.57
γ	1.63	1.69
e/a	1.47	1.46
δG (%)	22.52	21.20
δT_m (%)	17.11	17.33
Predicted elastic modulus	72.5	72.9

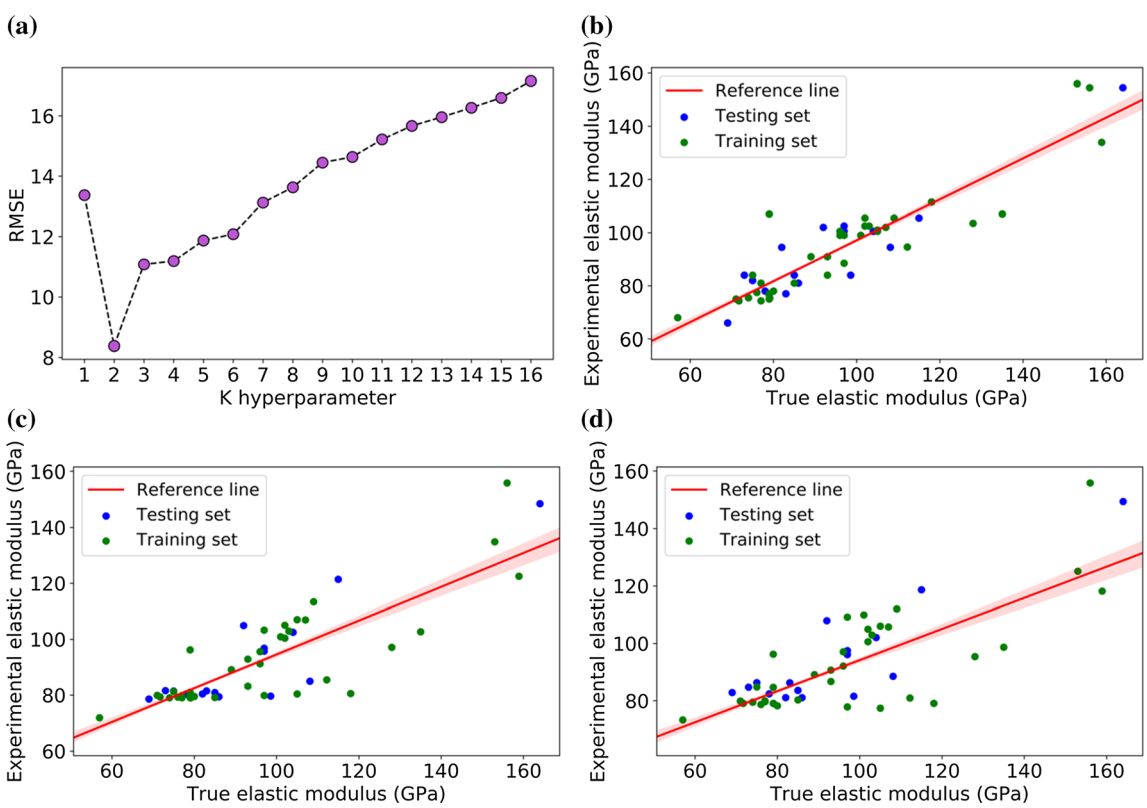


Figure 6 **a** The change in RMSE with increasing k hyperparameter for KNN algorithm. Performance of **b** KNN, **c** SVR.P and **d** SVR.L algorithms in predicting elastic modulus values as a function of measured elastic moduli of the alloys available in the testing set.

Validation experiments

The predicted compositions for Bio-HEA1 and Bio-HEA2 tabulated in Table 2 were prepared by vacuum arc melting under a Ti-gettered argon atmosphere on a water-cooled copper crucible. Raw metals of Ti, Ta, Hf, Nb with purity of 99.9% and Zr with purity of 99.2% were used in the production of Bio-HEA1 and Bio-HEA2. The produced alloys had a button shape with an intended mass of 20 g. To increase mixing and ensure chemical homogeneity, the button-shaped samples were flipped and re-melted at least 5 times. The button-shaped samples were then cut with wire electro-discharge machining (EDM) to obtain ten slices of each alloy with 2 mm thickness. Two slices from each sample were heat treated at 1000 °C for 12 h and then cooled down to room temperature in the furnace. (The heat-treated samples were named as Bio-HEA1-HT and Bio-HEA2-HT in the remainder of the paper.) To eliminate any contamination and surface residue, the samples were ground using SiC emery papers and polished with a 0.30 μm alumina slurry to obtain a mirror-like surface. To clean the bulk-samples, samples were immersed in and rinsed with de-ionized water. Then, the samples were exposed to immersion experiments in SBF for 28 days. The samples were kept in SBF, whose composition is presented in Table 3 [69], at a constant temperature of 37 °C throughout the whole experiment in an electronically controlled water bath. All HEA samples were held in separate sealed tubes, where the volume of the solution in each tube was calculated in accord with the G31 ASTM standards [70]. To measure the amount of mass gain or loss, the weights of the samples were recorded before and after immersion experiments.

Table 3 Composition for preparing 1000 mL of SBF solution utilized in the immersion experiments

Solution	Reagents	Amount (g/L)
SBF	NaCl	8.036
	NaHCO3	0.352
	KCl	0.225
	$K_2HPO4{\cdot}3H_2O$	0.23
	$MgCl_2{\cdot}6H_2O$	0.311
	1 M HCl	40 mL
	$CaCl_2{\cdot}2H_2O$	0.293
	Na_2SO_4	0.072
	TRIS	6.063
	1 M HCl	2 mL

The phase structures present in the samples were determined by X-ray diffraction (XRD) method utilizing a Bruker D2 Advanced X-ray diffractometer with Cu–Kα radiation source operated at 30 kV and 10 mA. The acquisition angle ranged from 20° to 100° with incremental steps of 0.02°. A scanning electron microscope (SEM) was employed to investigate the surface morphology of the samples. A depth-sensing nanoindentation device (Agilent G200) equipped with a Berkovich diamond indenter tip was employed to obtain the elastic modulus values of the samples from the loading and unloading curves by using the Oliver–Pharr method [72]. During the indentation experiments, a maximum load of 400 mN was applied at a constant loading rate of 40 mN/s in addition to a holding time of 5 s at maximum load. For each sample, a total of 15 measurements were performed, and the average values were presented with respective standard deviation. Furthermore, in order to quantify the quantity of ions released into the SBF solution during the immersion experiments, the remaining fluids of each static immersion tests were inspected employing an inductively coupled plasma-mass spectrometer (ICP-MS, Agilent 7700x).

Discussion of results

The XRD patterns of the untested Bio-HEA1, Bio-HEA1-HT, Bio-HEA2, and Bio-HEA2-HT are presented in Fig. 7. The results point out that all HEAs other than the Bio-HEA1-HT sample exhibited a single BCC phase with the lattice parameters determined as 3.486, 3.458 and 3.424 Å for Bio-HEA1, Bio-HEA2 and Bio-HEA2-HT, respectively, where the peak values were estimated utilizing Bragg's Law. The lattice parameter for the BCC phase of the Bio-HEA1-HT, on the other hand, was determined to be 3.452 Å. Although HEAs have a single crystal structure in as-cast condition, phase transformation and precipitation of intermetallic phases can be observed following heat treatment [73, 74]. For example, TiTaHfNbZr HEAs were shown to possess a single BCC crystal structure in as-cast condition and after heat treatments between 1000 and 1400 °C for 1 h [75]. Yet, they can produce different intermetallic phases (BCC Ta–Nb and/or HCP Zr–Hf rich precipitates) when heat treated between 500 and 900 °C for

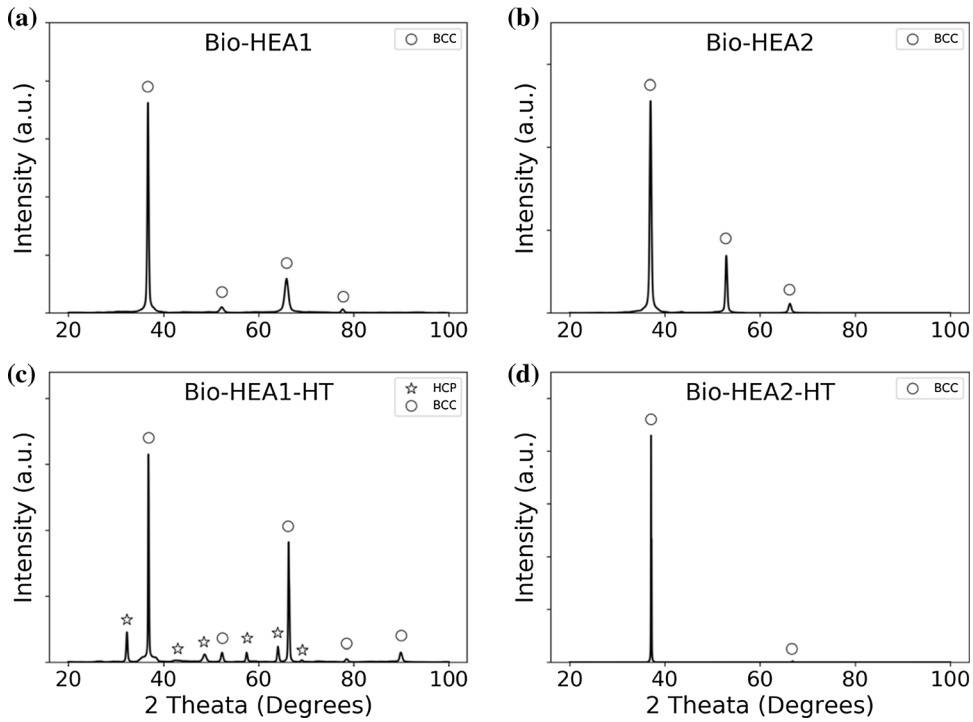


Figure 7 XRD results for **a** Bio-HEA1, **b** Bio-HEA2, **c** Bio-HEA1-HT, and **d** Bio-HEA2-HT. The star sign (☆) corresponds to $Hf_{0.43}Zr_{0.57}$ compound with hexagonal close-packed (HCP) structure [71].

1 h in addition to the main BCC matrix [75]. In Fig. 7c, the star sign (☆) corresponds to $Hf_{0.43}Zr_{0.57}$ compound with hexagonal close-packed (HCP) structure detected in the matrix of the Bio-HEA1-HT [71]. Even though Bio-HEA1-HT and Bio-HEA2-HT were exposed to the same heat treatment, the Bio-HEA2-HT features a single BCC structure, which is attributed to its smaller lattice parameter as opposed to other Bio-HEAs. Specifically, this might prevent Zr atoms, having the largest atomic radius in comparison to the other four elements, from diffusing within the matrix. In other words, the thermodynamic stability of Bio-HEA2-HT might be a consequence of high local lattice distortions resulting in sluggish atom diffusion [76].

The surface morphology of Bio-HEA1 (Fig. 8) and Bio-HEA2 (Fig. 9) was investigated by SEM prior to and following the static immersion experiments, revealing that both samples had a smooth surface prior to immersion experiments. However, following the immersion experiments, as opposed to Bio-HEA2, Bio-HEA1 had a highly corroded surface after immersion in SBF for 28 days, and a more locally intense corrosion behavior was prevalent as evidenced by the darker regions in Fig. 8f. This, moreover, may lead to severe pitting corrosion for long-term applications. On the other hand, based on Fig. 9b, d and f which show a few visible locally corroded regions, it can be argued that Bio-HEA2 is more resistant to SBF environment as compared to Bio-HEA1. The reason for Bio-HEA1 to have more visible locally intense corrosion behavior than Bio-HEA2 might be its higher Zr content exceeding the solid solubility level of Zr with other elements in the matrix [77]. It was reported that corrosion resistance of compositionally complex alloys first increased and then decreased with the increasing Zr content [77, 78]. The main cause of the decrease in corrosion resistance is that the excessive Zr enriching on the surface facilitates pitting corrosion, which, in turn, accelerates the corrosion [77]. In Fig. 10a and c, and Fig. 10b and d, the SEM micrographs of Bio-HEA1-HT and Bio-HEA2-HT samples prior to and following immersion in SBF for 28 days are presented, respectively. For both Bio-HEA1-HT and Bio-HEA2-HT, SBF environment was more deteriorating than it was for Bio-HEA1 and Bio-HEA2. This could be attributed to the formation of new phases in Bio-HEA1-HT and Bio-HEA2-HT as a result of heat treatment, which brings about a decrease in corrosion resistance due to chemical homogeneity in the matrix [79]. However, a detailed investigation of the passive oxide layer formation on the surface of the samples should be conducted by X-ray photoelectron spectroscopy (XPS) analysis to gain further insight and

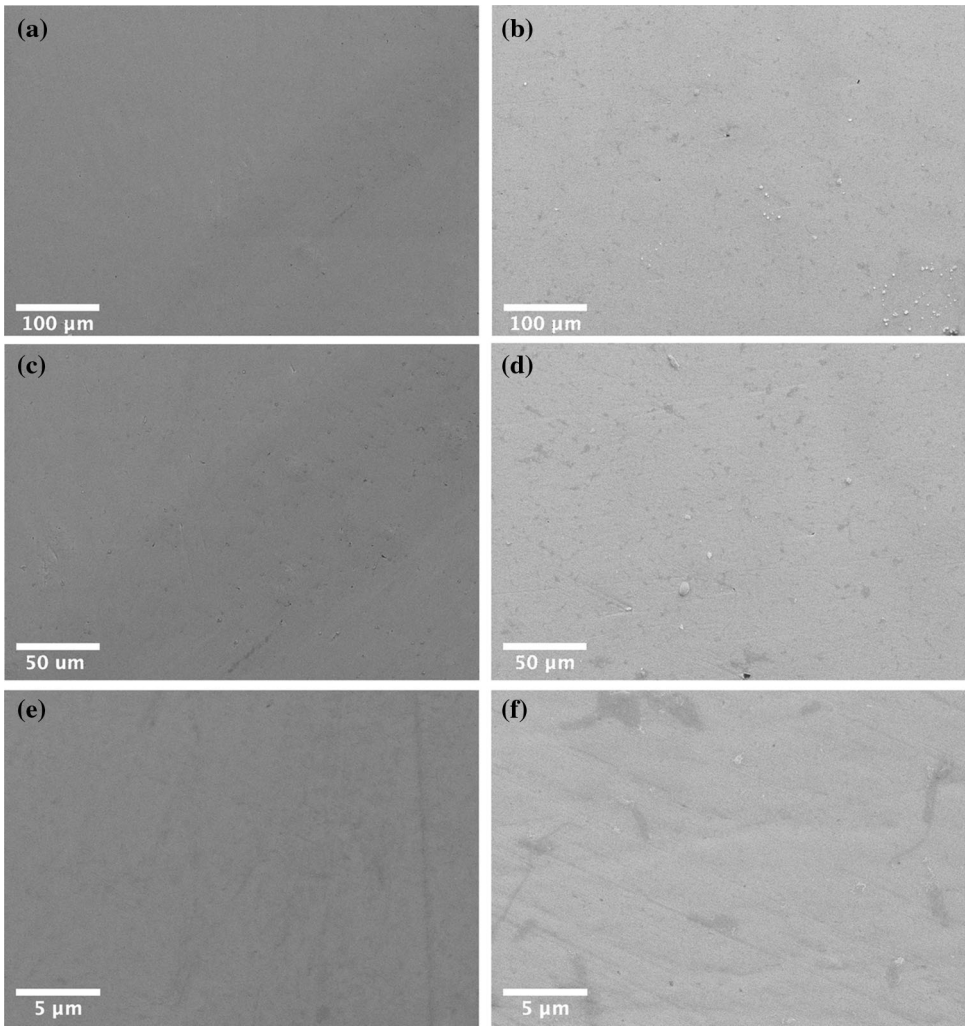


Figure 8 SEM images of the as-cast Bio-HEA1. **a**, **c**, **e** Prior to static immersion, with different magnifications showing a smooth surface. **b**, **d**, **f** the micrographs for Bio-HEA1 following 28 days immersion in SBF demonstrating the locally intense corrosion behavior (darker regions).

information about the surface layer properties of the samples immersed in SBF for 28 days.

As tabulated in Table 4, the quantity of total ion release from Bio-HEA1, Bio-HEA1-HT, Bio-HEA2, and Bio-HEA2-HT was very similar to each other following 28 days of immersion in SBF. However, it should be noted that only Ti ion concentrations released from all samples are demonstrated in Table 4 because of insignificant concentration of other elements. The main advantage of the HEAs studied in this work is that they do not include any toxic elements such as Al, Cr or V [80]. In particular, Ti is not a poisonous metallic element, and when present in human body in large doses, it passes through the intestines without being absorbed [80]. Based on a study conducted on the effect of Ti on osteoblast in rat calvaria cultures, it was reported that a Ti concentration higher than 10 ppm inhibits cell proliferation [81]. The concentrations of Ti ion released following a 28 days immersion in SBF from Bio-HEA1, Bio-HEA1-HT, Bio-HEA2, and Bio-HEA2-HT, as shown in Table 4, are well below the amounts that a human body can take in a day which is 800 µg/day [80].

Nanoindentation was utilized to measure the elastic modulus of each sample. Nanoindentation is a practical tool to assess the mechanical properties of materials. However, hardness and elastic modulus measurements calculated from loading and unloading curves by using the Oliver–Pharr method may be overestimated because of ignorance of pile-ups forming around the indent [82]. Therefore, extra pile-up area forming around the indenter was considered in this study to adjust the elastic modulus values measured by the nanoindentation experiments taking the pile-up effect into account. The method for correcting the measured elastic modulus values was described by Saha and Nix [83]. The method assumes

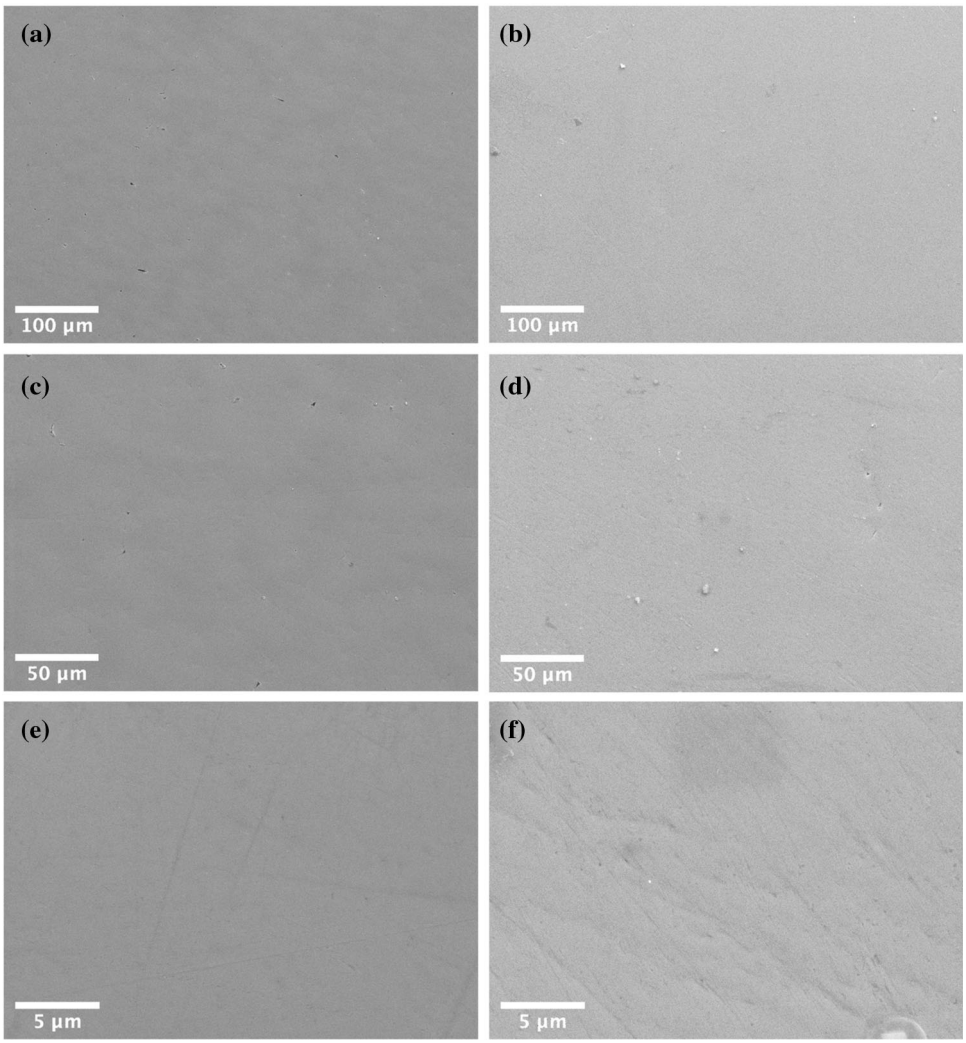


Figure 9 SEM micrographs of the as-cast Bio-HEA2. **a**, **c**, **e** Prior to immersion experiments, and **b**, **d**, **f** following immersion in SBF for 28 days with different magnifications.

that pile-ups form arcs around the edge of the indenter and the additional area due to pile-up effect is calculated utilizing geometry of the indenter [83]. In Table 5, the predicted and experimental elastic moduli values (after geometric correction) are presented for Bio-HEA1, Bio-HEA1-HT, Bio-HEA2 and Bio-HEA2-HT samples. The lowest elastic modulus measured is 83.5 ± 2.9 GPa, which belongs to Bio-HEA1 whose predicted elastic modulus was 72.5 GPa. The predicted and experimental elastic modulus results of Bio-HEA1 and Bio-HEA2 are close to each other verifying the effectiveness of ML methods employed in this work. The reason that the elastic modulus is higher for Bio-HEA1-HT and Bio-HEA2-HT as opposed to Bio-HEA1 and Bio-HEA2, respectively, is associated with the change in size, total volume and distribution of intermetallic phases in the matrix upon heat treatment, leading to alterations in the mechanical properties. For instance, the microhardness of $CoCrFeNiMnV_x$ HEAs was shown to first increase and then decrease with increasing and then decreasing amount of harder intermetallic phases concomitant with annealing temperature [84].

Conclusions

The results of the work presented herein demonstrate the capability of machine learning (ML) methods trained with a limited amount of experimental data to predict novel biomedical high entropy alloy (HEA) compositions with optimum elastic moduli for utility in orthopedic applications. During the training process of the ML algorithms, estimated empirical features were utilized in addition to compositions of the alloys in the dataset, revealing the importance of

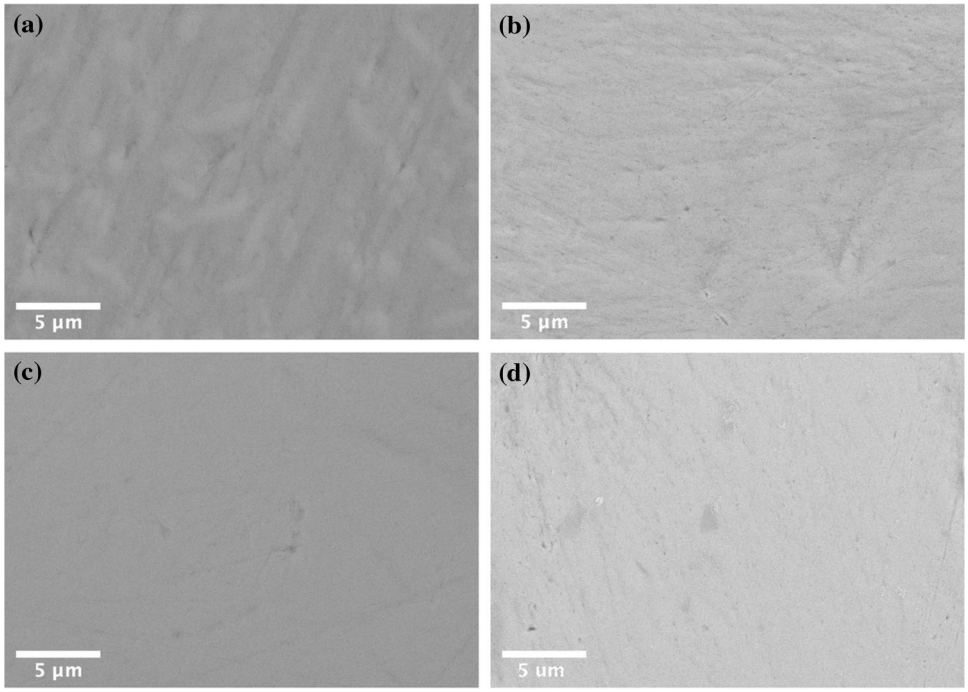


Figure 10 SEM images of Bio-HEA1-HT **a** before immersion and **b** after immersion in SBF for 28 days. SEM micrographs of Bio-HEA2-HT **c** before immersion and **d** after immersion in SBF for 28 days.

Table 4 Total ion concentrations released from Bio-HEA1, Bio-HEA1-HT, Bio-HEA2, and Bio-HEA2-HT following 28 days of immersion in SBF

Alloys	Quantity of total ion release in SBF (ppb)
Bio-HEA1	176.48 ± 2.50
Bio-HEA1-HT	183.45 ± 4.46
Bio-HEA2	181.20 ± 3.46
Bio-HEA2-HT	183.27 ± 3.36

Note that only the amount of Ti released into SBF is presented due to insignificant concentrations of Ta,Hf, Nb, and Zr

Table 5 Predicted and experimental elastic modulus values of Bio-HEA1, Bio-HEA1-HT, Bio-HEA2, and Bio-HEA2-HT obtained from nanoindentation experiments (Since heat-treatment parameters were not included in the dataset, predictions were made for compositions in as-cast condition)

	Predicted elastic modulus (GPa)	Experimental elastic modulus (GPa)
Bio-HEA1	72.5	83.5 ± 2.9
Bio-HEA1-HT	–	88.2 ± 2.6
Bio-HEA2	72.9	87.4 ± 2.2
Bio-HEA2-HT	–	90.5 ± 3.4

using intrinsic properties of the elements to increase the soundness of the model. The ML algorithm utilized in this work predicted two novel biomedical HEA compositions, namely $Ti_{23}Ta_{10}Hf_{27}Nb_{12}Zr_{28}$ and $Ti_{28}Ta_{10}Hf_{30}Nb_{14}Zr_{18}$, with elastic moduli of 83.5 ± 2.9 and 87.4 ± 2.2 GPa, respectively. The predicted elastic moduli of the new alloys were validated by nanoindentation experiments once the materials were manufactured. The samples were then subjected to immersion experiments in simulated body fluid (SBF) for 28 days to assess their biocompatibility and ion release behavior. The predicted alloys were also annealed in 1000 °C for 12 h to examine the changes in microstructure, and corresponding elastic moduli and ion release behavior. As a result, it was shown that heat treatment brought about formation of additional phases in the microstructure which showed that these alloys with a single BCC phase in the as-cast state can be used without heat treatment. Overall, the findings presented herein demonstrate that these materials have promising properties to be utilized as implant materials, but further investigation of their biocompatibility is necessary prior to their utility as orthopedic implant materials.

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